

동역학 숙제 #1

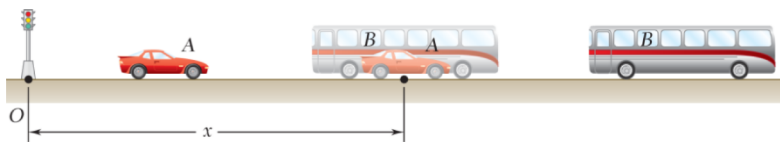
(김병일 교수)

(Due in Two Weeks)

문제 1)

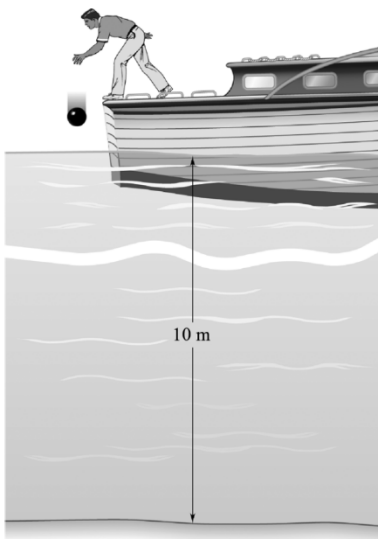
Starting from $x = 0$ with no initial velocity, a particle is given an acceleration $a = 0.1\sqrt{v^2 + 49}$, where a and v are expressed in m/s^2 and m/s , respectively. Determine (a) the position of the particle when $v = 24 \text{ m/s}$, (b) the speed of the particle when $x = 40 \text{ m}$.

문제 2)



Automobile A starts from O and accelerates at the constant rate of 0.75 m/s^2 . A short time later it is passed by bus B which is traveling in the opposite direction at a constant speed of 6 m/s . Knowing that bus B passes point O 20 s after automobile A started from there, determine when and where the vehicles passed each other.

문제 3)

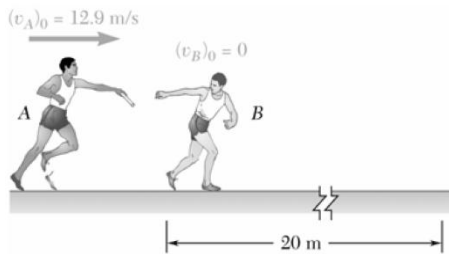


A bowling ball is dropped from a boat so that it strikes the surface of a lake with a speed of 8 m/s . Assuming the ball experiences a downward acceleration of $a = 3 - 0.1v^2$ when in the water, determine the velocity of the ball when it strikes the bottom of the lake.

문제 4)

The acceleration of a particle is directly proportional to the square of the time t . When $t = 0$, the particle is at $x = 24$ m. Knowing that at $t = 6$ s, $x = 96$ m and $v = 18$ m/s, express x and v in terms of t .

문제 5)



As relay runner A enters the 20-m-long exchange zone with a speed of 12.9 m/s, he begins to slow down. He hands the baton to runner B 1.82 s later as they leave the exchange zone with the same velocity. Determine (a) the uniform acceleration of each of the runners, (b) when runner B should begin to run.